

XAI study case on why customers churn 'Telco' company

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1. Introduction: problem description, motivation, stakeholders

1.1 Explaining the problem

Customer churn is one of the main issues for companies in the telecom industry. When people decide to cancel their contract, the company not only loses income but also ends up spending more money trying to replace those lost customers with new ones. Keeping the clients you already have is almost always cheaper and more efficient than finding others. For this reason, being able to understand why customers leave (and preventing it if it's possible) has a huge impact on a company. In this project we worked with a public dataset we found online about customers churn regarding the company called 'Telco'. From the first analysis, it came out that about 26.5% of their customers are classified as 'churned'. This is an high percentage and our goal is to understand how we can reduce this using XAI techniques. To reduce this churn rate we need to understand what drives people to abandon the service. Identifying the main reasons that make customers take this decision is the main point of this project.

1.2 Why we use Explainable AI

Traditional machine learning model (in particular the most powerful ones like XGBoost, the one we used in our project) are very useful to understand the customers that are going to churn the company. The problem is that their behavior is a black box model : they give you the prediction, but they don't explain the reason why they took a decision. For a team that needs to stop customers from leaving, like in our project, this is a problem, because we need to understand why a customer takes this decision in order to avoid his churn through different proposals. This is why Explainable AI is very important for our project. The aim of our work is that we want to change the focus from simply estimating the chance of churn to understanding what the company can do to avoid it. By using XAI methods, we want to understand the model's internal logic so that we can take decisions based on this.

1.3 Stakeholders and Project Goal

The company and group we chose for this work are the Telco Retention Team and the company's senior leadership. They need to understand why our XAI model takes his decisions in order to build strategies that will keep customers into the company.

In this report we are going to explain why having clear explanations is really important. First we look at the customers dataset we found online to try to understand which factors influence the most customer churn. Then we focus on one single customer who is considered at high risk from the model and examine the main reasons that are influencing him to leave the company. Finally we will highlight practical actions the company could take (based on the results of our XAI analysis) to keep the customers at risk. We will also see that investing in this actions would be financially useful, in order to show that even if we are spending money for our solutions, they will pay.

2. Data and Methods

2.1 Dataset and Preparation

For our project we found online the Telco Customer Churn dataset, which contains the data of 7,043 customer. We found a critical point at the beginning: only 26.5% of customers were classified as ‘churned’.

To prepare the data for the analysis, we first standardized them so everything could be analyzed. The churn was turned into a simple 1/0 (1 for customer at risk, 0 for customer not at risk) binary variable, missing values were estimated using the median, and the categorical features (that were words and not numbers) were converted into numbers by creating separate binary columns for each unique category.

2.2 Modeling Strategy and Evaluation

Our strategy is to compare a simple Logistic Regression (white box model) with a XGBoost model (black box model used for XAI).

Since the accuracy isn’t good with imbalanced data, we used on the ROC-AUC index as our metric to compare the models. We ensured data integrity by using a stratified split for training and testing.

We also verified that the XGBoost model, which we have selected for its higher ROC-AUC value respect to the Logistic Regression, wasn’t overfitted.

2.3 XAI Methodology

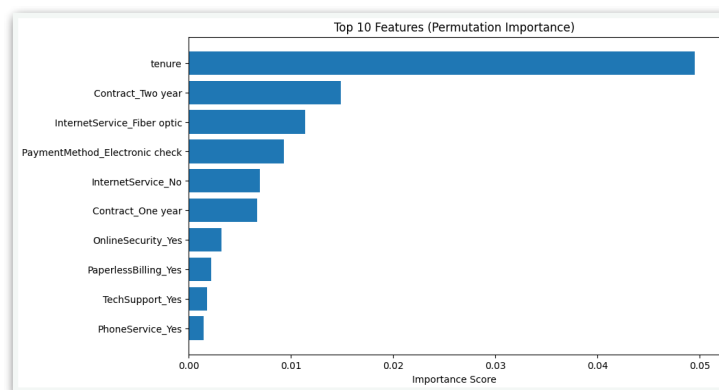


Fig 1. Global Feature Importance (Permutation). Tenure and Contract Type emerge as the dominant predictors across the model, aligning with domain intuition and confirming the model's robustness.

To understand how the XGBoost model works, we use a multi level XAI approach:

- Global Analysis: we used Permutation Importance plot and SHAP Summary plot to identify the main features why customers churn according to the whole dataset

- **Local Analysis:** here we used SHAP Force plots to understand feature influence on a single customer predicted as an high risk one
- **Robustness Checks:** last we performed stability checks comparing different results using different random seeds and reliability checks using random labels to verify that the predictions we have made were good

3. Results: Model Performance, Key Explanations, and Visualizations

3.1 Summary on the performance

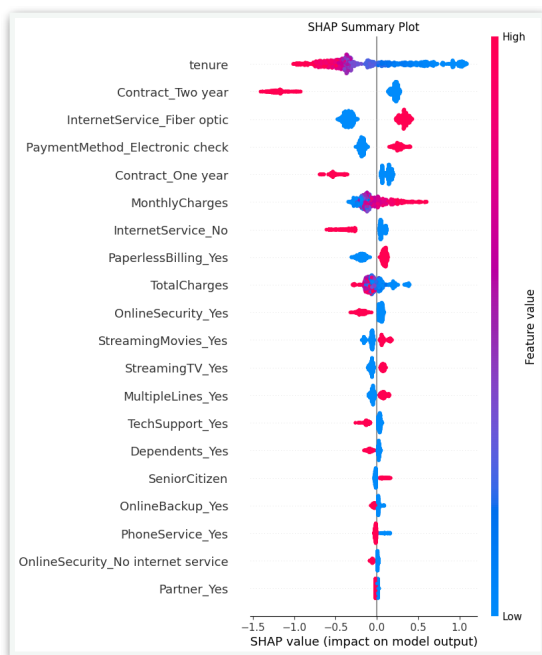
As we have described in Section 2, the XGBoost model was selected because it had better performances on predictions. In fact the model ROC-AUC index had a value of 0.8475, which is a good starting point for our analysis. This value indicates us that the model is robust, and the small gap between training and test index says us that the model is not overfitted and the new input are generalizable.

3.2 Global Drivers of Churn

To identify the main features influencing the model's predictions, we have examined the entire test set using Permutation Importance plot and SHAP Summary plot.

Permutation Importance identify us the three most important features: Tenure, Contract Type, and Internet Service. Together, these three variables have the strongest impact on model performance

SHAP plot gives us more informations, showing the direction and magnitude of each feature's effect:



1.Tenure: Customers with a low value of tenure (new customers) have high positive SHAP values, that influence the model's prediction to 'churn'. Instead customers with an high value on tenure have a negative influence on the decision

2.Contract Type: proposing a two-year contract is the best solution, because it generates negative SHAP values and so this reduces the probability of the model to take 'churn' as a decision

3.Internet Service: customers using fiber optic has an higher risk of churning, based on the graph

These results suggest us that proposing a new contract of 2 years is the most effective way the company has for convince customers not to leave the company.

Fig 2. SHAP Summary Plot. Features are ranked by importance. Red dots represent high feature values (e.g., Fiber Optic, Long-term contracts), while blue dots represent low values. The position on the x-axis shows the impact on churn probability: points to the right indicate higher risk.

3.3 Local diagnosis for a single high-risk customer

Now we want to analyze a single customer. To do this we use SHAP Force plot to obtain informations from a customer at high risk (customer number 1090), whose predicted churn value was of 99%.

The local analysis tells us that this high value of risk was because the customer's contract is a month-to-month one, which causes an increment in the risk of churn probability, and they use fiber optic, which is a feature that in this dataset is increasing the churn probability value for this dataset. Last, the customer has a short value of tenure, which means that he have been with the company for a short amount of time. These elements are increasing the churn probability of this customer to the high value of 99%.

The Force plot shows us that these features have a bigger effect than other things, like low total charges. Because of that, the model predicts a very high probability of the customer leaving the company.

4. Actions and Insights: model improvements and/or domain recommendations

4.1 Decisions based on obtained insights

Looking at the results of the XAI of Section 3, we notice something. Tenure is the most influent indicator on churn for a customer but what we want to know what we can actually do on his contract. Customers with monthly contract are the ones with the highest churn rate,so we need to find a solution to this problem.The ones with two years contract have very less risk than them to leave the company.

This means that the problem is mainly about the new customers subscribing short term (monthly) contracts. After we have predicted the churn we understand how to reduce it, in order to keep the customer.

4.2 Trying a what-if scenario

To understand the impact of an intervention based on the results we previously obtained, we simulated what would happen if all the customers identified as high risk switch their contract form the monthly one to a two years contract. So we made this change to the test set and the result was that from 285 customers initially expected to churn, 277 would remain if they have the long term contract. This means that 97% of the predicted churn customers would remain into the company if this happens, that is a very high (and unexpected) effect.We were very surprised from this results. So with this analysis we understood how much effective is if the company convince the customer to sign a longer contract then the monthly one.

The (hard) question now is how to convince the customer to sign a two year contract.

4.3 Suggestion based on the model results

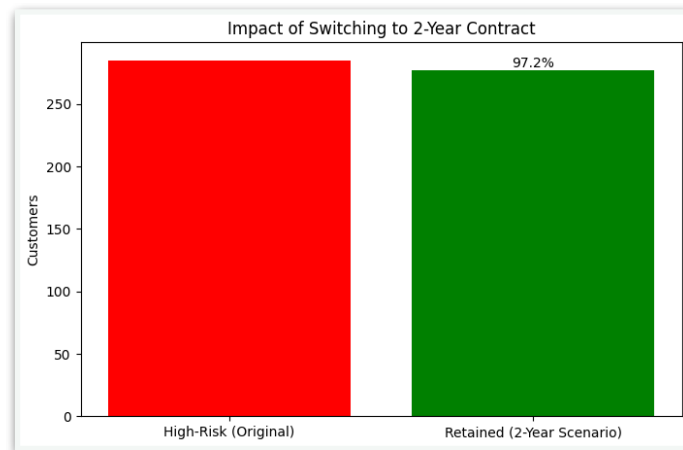


Fig 3. Impact of Strategic Intervention. Simulating a switch to a two-year contract for high-risk customers resulted in a 97.2% retention rate, validating the effectiveness of long-term contract incentives.

So from this analysis we understood that if we want to keep the customers who has an high churn value, we need to convince customers with monthly contract to sign a long term one. Some simple ideas to do this (just proposal ideas we came out with) are: a discount or a deal to make it convenient to switch to this type of contract. This would be for sure very effective based on the fact that if we convince them, the churn rate would decrease for the 97%. In other words, this approach would solve the problem (that we have identified), which isn't the services offer itself but the high risk of abandon the company from customers with monthly contracts (again, the main feature affecting customers churn rate).

What makes the idea even stronger is that longer contracts give the company a more stable and periodic revenue, so any incentive offered would be the best solution to take for the company. At this point, it's not just about analyzing the churn rate. The XAI results give us very useful indication on how to solve this problem, and make it obvious where the solution we found should go in order to keep the customers in long terms.

5. Discussion: limitations, risks, and ideas for future work

5.1 Validating our suggestions

At the end the analysis has shown that the decision to keep clients is very clear and completely identified. If we want to keep these customers into the company, we need to understand how we can make customers sign a two years contract, in order to obtain this we should offer something that is advantageous enough for people currently on month plans to consider changing. To do that we could use a discount, some extra services, or some more benefits, anyway the goal is to give them a reason to stay. And the analysis says us that this effort would pay off, because the model indicates that convincing them to take the longer contract could convince almost all of the customers who are currently at risk not to abandon the company.

Our analysis is made so that it highlights the main problem; it isn't the fact that customers are disappointed with the service, but it's the absence of a constraint that keep the client into the company; at the moment, the customer can go away whenever he wants. Investing in obtaining a longer agreement help to avoid this problem, that gives way more stable revenue to the company, which means that every incentive offered to the customers converts in a more profitable investment in long terms. At the end, XAI results are very practical : they give us a very clear direction to follow and to invest on. This result is very practical and it's what we needed rather than some other dataset of numbers that need to be interpret.

5.2 Differences between correlation and the real intervention

The analysis shows that customers with long term contracts churn less than the ones with monthly contracts. This is an important signal, but we have to interpret this evidence very carefully. The model shows a strong pattern. Customers who choose the two years contract are usually already comfortable with the service, and this satisfaction is the main reason why they stay longer. In other words: the long contract is often a consequence. They see that the product that the company offers is valid and so they decide to sign a long agreement.

For this reason, the 97% reduction on churn rate we obtained from switching high risk users to two years plans should be interpreted as an maximum result we can obtain rather than a real prevision. Customers react in different ways that the model cannot predict. For example,competitors could offer a more convenient promotion; a technical problem could appear after the contract change; and also many people don't want to sign long contracts in order to not being constrained for two years, even if the offer is good for them. All these behaviours aren't visible in the dataset, and the model don't takes into account for them.

What the data analysis is actually telling us is that the contract structure is a strong factor to keep customers: shifting a certain account of customers to longer contracts can reduce churn, but only if the customers agrees the change in the contract and sees it as convenient. The intervention works we should make is convenient when the customer wants to stay, not when the company tries to force customers to stay.

5.3 Final synthesis and directions for future work

This project was useful to highlight the value of using XAI as a practical tool to take decisions. By understanding how the model formulates its predictions, we identified a specific structural weakness in the customers agreements : monthly contracts expose the company to a much higher churn risk. Seeing this fact using SHAP made it clear that the contract type is not just a relevant feature,it is the main reason why many retention decisions are taken form customers leaving.

For this reason, several improvements emerge. One first intervention should be refining how incentives are offered to the client. The actual simulation assumes that switching to a two years contract changes totally the prediction, but a more realistic approach should also study which promotions and incentives actually make the customer agree in practice. Another useful analysis should be the fair analysis: before taking any decisions simulating the customers, the company should verify that the model does not implicitly favour certain groups over others. A final insight should be the model simplification. Some features add very little predictive value, and removing them could make the model easier to communicate and more efficient without losing accuracy; we should just identify the less important features and remove them.

Overall, the project demonstrated us that combining predictive model with explanation tools helps move from some assumptions about the churn rate to a concrete intervention. It identifies where we should take action to solve the problem and gives the team a clear direction to follow.

Dataset source: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>